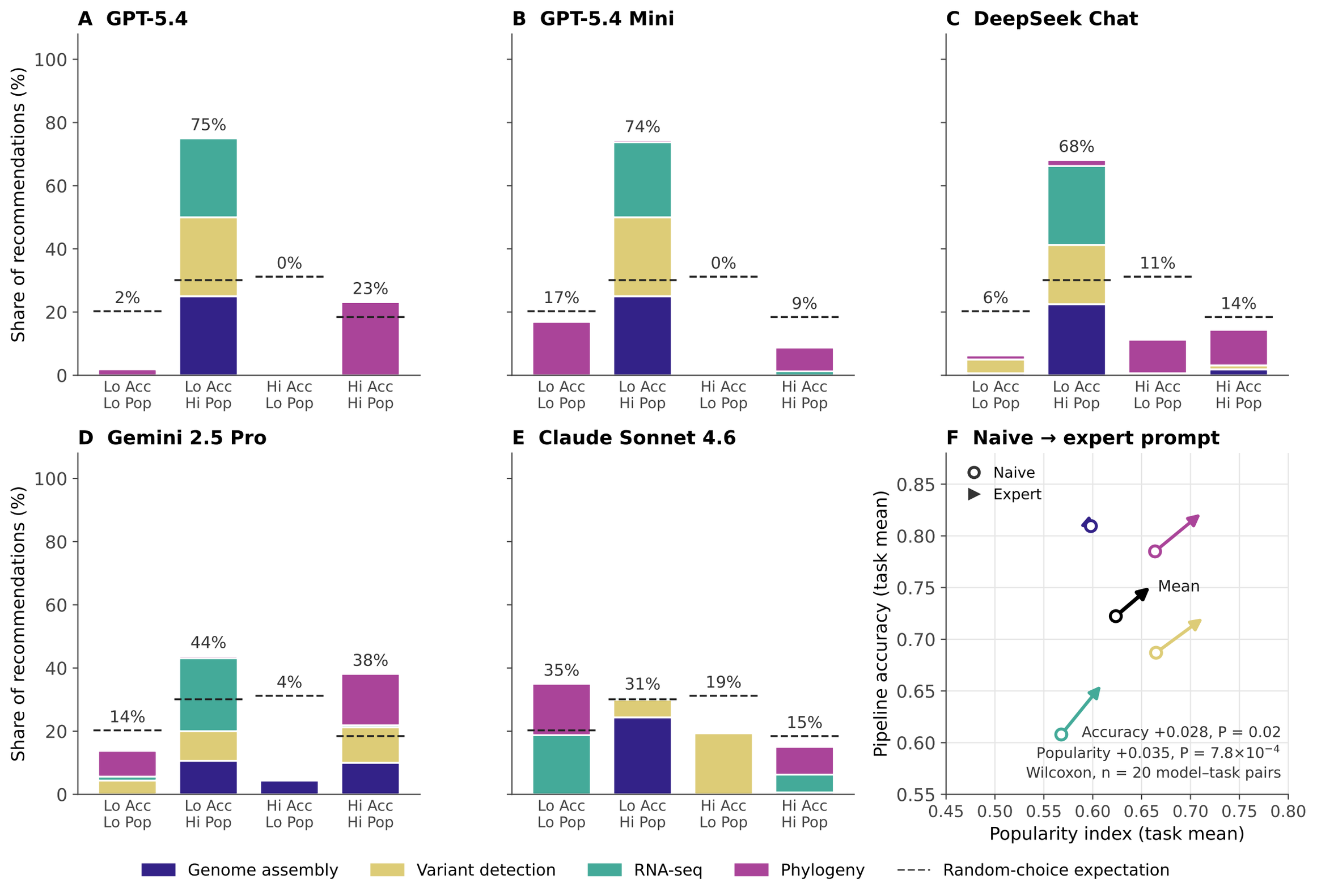
A GPT-5.4
B GPT-5.4 Mini
C DeepSeek Chat
D Gemini 2.5 Pro
E Claude Sonnet 4.6
F Naive → expert prompt
Share of recommendations (%)
0
20
40
60
80
100
2%
75%
0%
23%
17%
74%
0%
9%
6%
68%
11%
14%
14%
44%
4%
38%
35%
31%
19%
15%
Lo Acc Lo Pop
Lo Acc Hi Pop
Hi Acc Lo Pop
Hi Acc Hi Pop
Naive
Expert
Mean
Pipeline accuracy (task mean)
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.45
0.50
0.55
0.60
0.65
0.70
0.75
0.80
Popularity index (task mean)
Accuracy +0.028, P = 0.02
Popularity +0.035, P = 7.8×10⁻⁴
Wilcoxon, n = 20 model–task pairs
Genome assembly
Variant detection
RNA-seq
Phylogeny
Random-choice expectation